MIL-B-20295
21 November 1951

MILITARY SPECIFICATION

BLANKS, CENTRIFUGAL CAST, ROTATING BAND

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army 50-27-3
19 May 1948

This specification consists of this cover sheet and Specification 50-27-3, dated 19 May 1948, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer .

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U. S. ARMY
SPECIFICATION

No. 50-27-3
19 May 1948

BLANKS, CENTRIFUGAL CAST, ROTATING BAND

A. APPLICABLE SPECIFICATION.

A-1. Federal Specification QQ-M-151, Metals: General Specification for Inspection of, of the issue in effect on date of invitation for bids, forms a part of this specification.

B. CLASS AND CONDITION.

B-1. Class. This specification covers one class of rotating band blank produced by the centrifugal casting process.

B-2. Condition.-Rotating band blanks shall be annealed, if necessary, to meet physical requirements.

C. MATERIAL AND WORKMANSHIP.

C-1. Material.-Raw materials used shall be such as to produce rotating band blanks meeting the requirements of this specification.

C-2. Workmanship.-

C-2a. The blanks shall be in the form of a solid ring.

C-2b. The castings shall be of uniform quality and condition, free from injurious blow holes, porosity, hard spots, shrinkage defects, cracks, and other harmful defects.

C-2c. All blanks shall be round and of uniform diameter, thickness, and width, and shall be free from scale, chips, oil, grease, and dirt.

C-2d. Annealed blanks, unless bright-annealed, shall be pickled, washed, and dried.

D. GENERAL REQUIREMENTS.

D-1. See section E.

E. DETAIL REQUIREMENTS.

E-1. Chemical composition.-

E-1a. The chemical composition shall meet the requirements shown in Table I.

TABLE I - CHEMICAL REQUIREMENTS

Copper (a), percent	89.0 - 93.0
Iron, percent, maximum	0.050
Lead, percent, maximum	0.050
Total elements other than copper and zinc, percent maximum	0.125
Zinc	Remainder

(a) - Silver to be counted as copper.

E-1b. An analysis of each melt shall be furnished by the contractor, showing the percentages of the elements designated in Table I.

E-2. Tensile properties.-The tensile properties of the annealed blanks shall meet the requirements shown in Table II.

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TABLE II - TENSILE REQUIREMENTS

Tensile Strength, psi, max.	Elongation in 2 inches, percent, min.	Angle of bend (Degrees)
38,000	35	180

E-3. Bend test. Annealed blanks shall withstand the bend test prescribed in paragraph F-6, without cracking on the outside of the bent portion.

E-4. Brinell hardness (500 kg. load 10 mm Ball) shall be not greater than 65.

E-5. Dimensions and permissible variations shall be as specified in the contract or purchase order or on applicable drawings.

F. METHODS OF INSPECTION AND TESTS.

F-1. Size of lots.- A lot shall consist of not more than 20,000 blanks of the same condition and size, manufactured by the same process.

F-2. Dimensions. At least five blanks from each lot shall be carefully checked for compliance with the dimensional requirements.

F-3. Sample for government analysis. A sample for Government analysis shall be taken from each lot. Each sample shall be composed of approximately equal parts of drillings or millings taken from two or more blanks selected by the inspector, and shall weigh not less than 5 ounces.

F-4. Samples for physical tests.

F-4a. A sufficient number of bands shall be selected by the inspector from each lot to provide for two tension tests, two bend tests, and at least two Brinell hardness tests for each 1,000 blanks in the lot. Whenever practicable, specimens for both the tension test and the bend test may be taken from the same blanks.

F-5. Tension test specimens.

F-5a. Tension test specimens shall be cut circumferentially from blanks. The blank, or section thereof, shall be straightened cold by pressure, without hammering. Test specimens shall be in accordance with type 5 specimen, Federal Specification QQ-M-151. The thickness of test specimens shall be the full thickness of the blank, provided that a small amount of metal shall be removed by machining when necessary to obtain a suitable test specimen.

F-6. Bend test.-

F-6a. Bend test specimens shall consist of a section of the full width and thickness of the blank, with the subtended arc not less than one-third nor more than one-half of the circumference of the blank.

F-6b. The specimen shall be bent cold, by pressure without hammering, in the direction of curvature of the blank, until it tightly incloses 180° of a rod equal in diameter to the thickness of the blank.

F-7. Retests. Retests may be made in accordance with Federal Specification QQ-M-151.

G. PACKING AND MARKING.

G-1. Packing.

G-1a. Blanks shall be separated according to condition and size when packed for shipment.

G-1b. Packing shall be such as to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery.

G-2. Marking. Each shipment shall be legibly and indelibly marked to show the size, lot number, gross and net weight, quantity, name of the contractor, and the contract or purchase order number.

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H. NOTES.

H-1. Rotating band blanks covered by this specification are intended primarily for projectiles 155 mm and larger.

H-2. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

NOTICE: This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government, and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

Note. Copies of this specification may be obtained from the Office, Chief of Ordnance, Department of the Army, Washington 25, D.C.

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